Punit Daga

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Skills

Languages	**English (Native)**, **Russian (Proficient)**, **Hindi (Proficient)**
Programming Languages	**C**, **C++**, **Assembly**, **VHDL**, Python, PowerShell, **JavaScript (Node.js/ES6+)**,**TypeScript**, **HTML5**, **CSS3**, **Bash/Shell**
Test Automation & QA	**Playwright**, **Appium (iOS & Android)**, **End-to-End (E2E) Testing**, **Page Object Model (POM)**, **Cross-Browser Testing**, **Mobile Test Automation**, **API Contract Testing**, **Regression & Smoke Testing**, **Anti-Flake Engineering**, **Root Cause Analysis (RCA)**
Embedded Systems & RTOS	**RTOS development**, **real-time systems design**, **memory management**, **driver development**
Cloud & DevOps	**Firebase**, **Azure**, **Azure DevOps**, **Google Cloud Platform (GCP)**, **AWS (Amazon Web Services)**, **Jenkins**, **Docker**, **GitHub Actions**, **CI/CD Pipelines**, **Network Diagnostics (BGP/Traceroute)**, **OpenVPN**
AI & Agentic Testing	**Agentic QA Evaluation**, **Claude (Projects)**, **GPT-4**, **Gemini**, **LLM Benchmarking**, **Prompt Engineering**
Databases & Storage	**PostgreSQL**, **SQL**, NoSQL (FirebaseDB, MongoDB)
Security & Cryptography	**OAuth**, **Network Security**, **Data Encryption**, **Application Security Controls**, Post Quantum Encryption (PQE), Zero Knowledge Proofs (ZKPs)
Tools & Technologies	**Git**, Docker, **JSON**, VS Code, Linux, **Jira**, **Linear**, **Chrome DevTools**, **Lucidchart**, **Monaco Editor**, **Notion**, **Tampermonkey**

Education

University of Ottawa | Washington Accord Accredited Degree *Ottawa, Canada*

BASc in Computer Engineering with a focus on hardware design and networking. *September 2018 - May 2024*

- **Embedded Systems & RTOS:** Developed expertise in **real-time operating systems (RTOS)** development, **real-time systems design**, **concurrency and synchronization**, **memory management**, and **device driver development** through rigorous coursework and hands-on projects.
- **Hardware Design:** Acquired skills in **digital computer design**, **CPU design & control**, **memory unit design**, **FPGA design**, **VHDL/HDL design**, and **embedded systems**, providing a solid foundation for hardware-level considerations.
- **Software Development:** Built a strong foundation in **software engineering** principles, **object-oriented programming (OOP)**, **data structures**, **concurrency and synchronization**, **memory management**, and the **software development lifecycle** (SDLC) to create efficient and scalable software solutions.
- **Networking:** Gained comprehensive knowledge of **network protocols** (TCP/IP, UDP), **IPv4/IPv6 addressing**, **network security protocols**, **routing algorithms**, **communication protocols** (UART/RS232, RS485, I2C, SPI, BLE), essential for developing and maintaining secure and robust networked systems.
- **Security & Cryptography:** Developed a deep understanding of **network security protocols**, **encryption algorithms**, and **data protection mechanisms** through coursework and practical projects.

Experience

Work | QA Wolf *Remote*

Software Development Engineer in Test (SDET) / QAE II *August 2024 - September 2026*

- Promoted to **Level II in 11 months**; architected modular **Page Object Model (POM)** class hierarchies and design patterns, leading an engineering-wide migration from procedural scripts to maintainable OOP codebases.
- Engineered developer tooling and internal utilities, including a Monaco-based IDE plugin (**qawolf-ide-themer**) and an automated media-export service to reliably process and serialize client video artifacts under strict SLAs.
- Diagnosed and resolved complex distributed systems and network bottlenecks, isolating upstream **BGP routing failures** on test runners via traceroute/RPKI analysis and optimizing **WebRTC stream latency** via selective **OpenVPN split-tunneling**.
- Optimized runtime performance and determinism across **1,500+ tests** in **Playwright** and **Appium** (JavaScript/TypeScript), eliminating timeout-heavy polling loops in favor of deterministic DOM assertions and native Android OS event listeners.
- Integrated scalable pre-merge validation pipelines across **GitHub Actions** and **Azure DevOps**; developed LLM orchestration workflows (Claude, GPT-4, Gemini) that cut automated workflow generation from 3 days to **15 minutes**.

Project | ScrapYard - Automated Trash Sorting System *Ottawa, Canada*

Fullstack Developer *September 2022 - April 2023*

- Created a sustainability-focused **garbage and recycling sorter prototype** for use in public spaces
- Developed **web app**, **Express.js server**, **PostgreSQL database**, **OAuth authentication**, React Leaflet maps, and integrated **Google Cloud (GCP) APIs/Firebase**
- Contributed to the development of **AI-powered computer vision** for object recognition and **automated control systems** for sorting

Project | IDP Solar Grid Simulator

Ottawa, Canada

Project Lead

January 2023 - April 2023

- Developed cyclic scheduler on **BeagleBone Black** using **SYS/BIOS** for real-time task management
- Implemented **MVC** pattern for terminal-based user interface on BeagleBone Black
- Managed continuous data acquisition from **SOURCE** and **LOAD** ports with BeagleBone Black's **ADC**
- Conducted **RMS voltage**, current, and power signal analysis on BeagleBone Black
- Integrated **SPI**, **ADC**, **GPIO**, and **UART** interfaces on **AM335x microcontroller**
- Utilized SYS/BIOS for **multi-threading** and task execution on BeagleBone Black
- Conducted detailed functional and performance **analysis** for **system optimization**

Work | Software Developer

Toronto, Canada

Lonehaven

January 2022 - December 2022

- Developed **React.js** marketing websites and **Shopify** e-commerce stores for clients, serving 20,000+ customers
- Utilized **Shopify API**, **React.js**, **Angular**, and **AWS** for deployment and management
- Led the development of the company website and **managed** a team of 5 student developers on various projects for 6 months

Project | CEG3155 UART Traffic Light Controller

Ottawa, Canada

Hardware Designer

September 2021 - December 2021

- Developed a **VHDL**-based **traffic light control system** using **Finite State Machine (FSM)** with one-hot state assignment and Sunggu Lee class equivalence method for state minimization
- Assembled a **prototype** for **CMOS** to **RS-232** signal conversion using a breadboard, capacitors, and MAX232 dual transmitter/receiver
- Created a **micro-controller** simulated FSM interfacing with **UART** in structural VHDL
- Implemented **UART's** Transmit and Receive **Shift Registers** using 8-bit **PIPO** and **SIPO** shift registers with D Flip-Flops
- Designed a **UART Transmitter** with TSR, a 16-bit counter, and D Flip-Flops, enabling **ASCII character transmission** to an Altera DE2-115 **FPGA** via the MAX232 Tx connection

Work | Software Engineer

Ottawa, Canada

Ottawa Department of Medicine

August 2020 - April 2021

- Collaborated with Microsoft Engineers to develop a **React.js web application** with a questionnaire and **PDF rendering** functionality for internal use
- Built the UI using MUI and implemented PDF rendering using react-pdf
- Deployed and managed the application on a **Node.js server** within the **Azure** cloud environment
- Utilized **Azure DevOps** for project management and **CI/CD pipelines**

Work | Network Technician

Ottawa, Canada

Smart Living Properties

July 2019 - May 2020

- Provided on-site **network troubleshooting** and support for home **internet** services
- Resolved **security incidents** (Gozi spyware infections) on customer networks
- Deployed and configured enterprise-grade **network solutions** using **Cisco Meraki** equipment

Project | Emergency Call Bell System

Ottawa, Canada

Fullstack Developer

January 2020 - April 2020

- Implemented **back-end services** for a **real-time** emergency response system, emphasizing **network communication** and **data processing**
- Developed and integrated **networking features** for fall detection and automated emergency response, ensuring system **reliability** and **efficiency**
- Collaborated in a multidisciplinary team, contributing expertise in **networking**, system analysis, and **embedded systems programming**